CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Python Data Preparation | SQL Analysis | Power BI Dashboard

Companion files: Jupyter notebook (.ipynb) and SQL queries (.sql)

1. Project Overview

This project examines 3,900 customer purchase records to understand spending patterns, product preferences, subscription behavior, and differences across customer groups. The work follows a complete analytics process: explore and clean the data in Python, use SQL to answer business questions, and present the main patterns in an interactive Power BI dashboard. The goal is not just to calculate totals, but to turn those results into practical decisions about marketing, customer retention, and product strategy.

2. Dataset Summary

The original dataset contains 3,900 rows and 18 columns. It includes customer information such as age, gender, location, and subscription status; purchase information such as item, category, amount, season, size, and color; and behavior-related fields such as discounts, previous purchases, shopping frequency, shipping method, and review ratings.

Initial data quality issue: 37 review ratings were missing. These were handled during cleaning rather than dropping the affected records.

3. Data Preparation and Exploration in Python

Initial exploration

I loaded the CSV into a pandas DataFrame and used `head()`, `info()`, `describe()`, and null-value counts to inspect the data. This helped me check column types, understand the ranges of numerical values, and identify missing information before running any analysis.

Handling missing review ratings

I filled missing review ratings with the median rating for the corresponding product category. Using the category median preserves the records while keeping the replacement values tied to similar products. It is also less affected by unusually high or low ratings than a mean.

Standardizing and creating fields

I converted column names to lowercase snake_case and renamed the purchase amount field to `purchase_amount`. I then created `age_group` using four quantile-based age bands: Young Adult, Adult, Middle-aged, and Senior. I also mapped the text-based purchase frequency labels to approximate day counts in `purchase_frequency_days`, making frequency easier to compare numerically.

Finally, I checked whether `discount_applied` and `promo_code_used` contained the same values. The notebook confirmed they matched throughout the dataset, so I removed the redundant promo code field to avoid duplicate information.

Database preparation

The notebook includes database connection examples for PostgreSQL, MySQL, and SQL Server. Its Python workflow prepares the cleaned DataFrame for loading into a relational table. The supplied SQL analysis script uses PostgreSQL-style syntax, including a numeric cast in the review-rating calculation. The Power BI file uses a customer shopping behavior table as its data source.

4. SQL Analysis: Business Questions and Findings

I used SQL aggregations, conditional filters, a subquery, a common table expression (CTE), and a window function to investigate ten questions. The findings below use the results shown in the original report where those results are available. The supplied SQL script follows the same ten-question structure.

1. Revenue by gender

Approach: Grouped purchases by gender and summed purchase_amount to compare revenue contributions.

Finding and interpretation: Male customers generated $157,890 and female customers generated $75,191. This is a revenue comparison, not proof that individual male customers spend more per purchase.

2. High-spending customers using discounts

Approach: Filtered records with a discount and a purchase amount at least as high as the overall average, calculated with a subquery.

Finding and interpretation: The original result contains 839 qualifying purchase records. These transactions suggest discounts are not limited to low-value purchases; profitability would require margin data.

3. Top five products by review rating

Approach: Calculated average review rating by product, rounded the result, sorted descending, and kept the top five.

Finding and interpretation: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), and Skirt (3.78) led the original ranking. Ratings show satisfaction, not necessarily sales volume.

4. Standard versus Express shipping

Approach: Filtered the two shipping types and compared their average purchase amounts using AVG().

Finding and interpretation: Express orders averaged $60.48 compared with $58.46 for Standard orders. The $2.02 difference is small and does not establish that shipping choice causes higher spending.

5. Subscribers versus non-subscribers

Approach: Grouped by subscription status and calculated record counts, average purchase amount, and total revenue. ORDER BY total_revenue DESC, avg_spend DESC displays the highest-revenue group first and uses average spending to break revenue ties; it changes presentation, not the underlying figures.

Finding and interpretation: The original report shows 1,053 subscribed records with $59.49 average spend and $62,645 revenue, versus 2,847 non-subscribed records with $59.87 average spend and $170,436 revenue. The groups have similar average purchase amounts.

6. Products most frequently purchased with discounts

Approach: Used a CASE expression to count discounted purchases and divided by all purchases for each product to calculate a percentage.

Finding and interpretation: Hat (50.00%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%) had the highest discount rates in the original output. This measures discount usage, not whether discounts increased demand.

7. Customer segmentation

Approach: Used a CTE with CASE to label previous_purchases = 1 as New, 2 through 10 as Returning, and higher values as Loyal; then counted each group.

Finding and interpretation: The original breakdown was Loyal 3,116, Returning 701, and New 83. These labels are based on the previous-purchases field, not a full longitudinal record of customer activity.

8. Top products within each category

Approach: Counted orders by category and product, then used RANK() partitioned by category to rank products against others in the same category. Unlike ROW_NUMBER(), RANK() preserves ties and skips subsequent positions when ties occur.

Finding and interpretation: The updated output shows Accessories: Jewelry (171), Sunglasses (161), Belt (161); Clothing: Blouse (171), Pants (171), Shirt (169); Footwear: Sandals (160), Shoes (150), Sneakers (145); Outerwear: Jacket (163), Coat (161). Blouse and Pants share rank 1, so Shirt is rank 3; Sunglasses and Belt share rank 2. Because Outerwear has only two products, only two rows appear for that category.

9. Repeat purchases and subscriptions

Approach: Filtered customers with more than five previous purchases and counted their subscription statuses.

Finding and interpretation: Among qualifying records, 958 were subscribed and 2,518 were not. Because the query reports counts rather than within-group subscription rates, it does not by itself show whether repeat buyers are more likely to subscribe.

10. Revenue by age group

Approach: Summed purchase_amount for each age_group and sorted by revenue contribution.

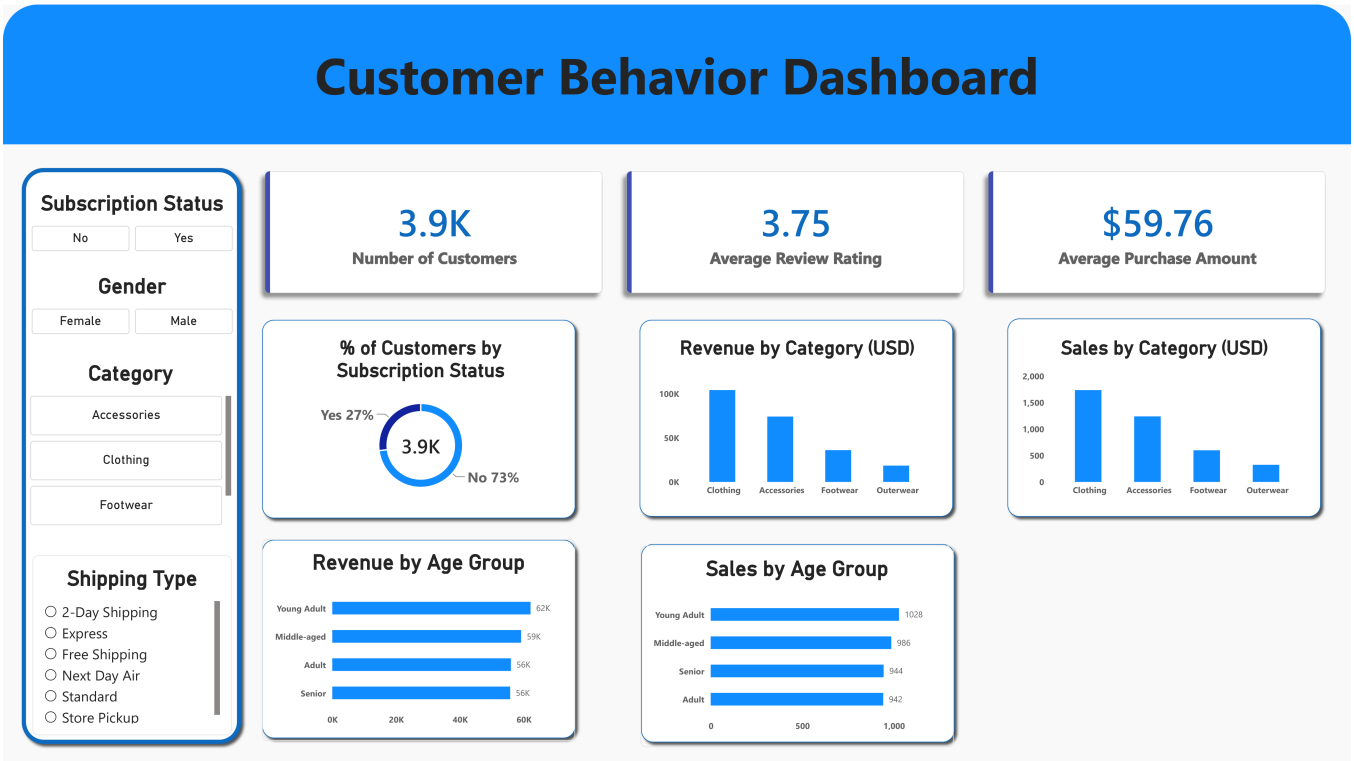
Finding and interpretation: Young Adult led at $62,143, followed by Middle-aged ($59,197), Adult ($55,978), and Senior ($55,763) in the original output. These totals reflect group size as well as spending behavior.

5. Power BI Dashboard

I built an interactive Power BI dashboard to make the main results easier to explore. The exported dashboard contains KPI cards, a subscription-status donut chart, category-level column charts, age-group bar charts, and slicers for

subscription status, gender, category, and shipping type. Together, these visuals let someone move from an overall summary to a more focused comparison without rewriting SQL queries.

The dashboard below is the updated version exported from Power BI. It replaces the earlier pink dashboard shown in the original report.



The KPI cards summarize the number of customers, average purchase amount, and average review rating. The donut chart shows the subscriber mix. Category charts compare revenue and sales volume, while the age-group charts show how revenue and purchase counts vary across age bands. Slicers let the viewer filter the dashboard and see how the measures change for selected customer groups.

Dashboard interpretation: Revenue and transaction count should be read together. A category can generate more revenue because it sells more items, because its average purchase amount is higher, or both. Similarly, a large customer segment can contribute more total revenue without having the highest average spend.

6. Business Recommendations

Improve subscription value

Subscribers and non-subscribers had similar average purchase amounts in the original results. Instead of assuming subscribers spend more, the business could test targeted membership benefits and track whether they improve retention or purchase frequency.

Reward repeat customers

The customer segmentation query identifies a large loyal group based on prior purchases. A loyalty program or personalized offers could help retain those shoppers, while a separate onboarding campaign could encourage newer

customers to return.

Review discount strategy

Some products appear in discounted purchases nearly half the time. The business should compare promotional performance with profit margins before expanding discounts, since higher sales do not automatically mean higher profit.

Use product and demographic insights carefully

Highly rated products are reasonable candidates for promotional placement, and revenue by age group can guide campaign testing. However, these comparisons should be paired with purchase counts and average spending so that larger groups are not mistaken for more valuable customers on an individual basis.

7. Limitations and Next Steps

The dataset represents a snapshot of purchase records, so the analysis describes associations rather than proving cause and effect. It also lacks the product costs and profit margins needed to evaluate discount profitability. Customer counts in the queries are based on customer_id rows; unique-customer analysis would require checking whether IDs repeat. As a next step, I would validate the refreshed SQL outputs against the Power BI measures and consider adding subscription conversion rates, average spend per customer, and margin-based promotional metrics.

8. Conclusion

This project gave me experience working through an end-to-end data analysis process. I used Python to prepare a consistent dataset, SQL to answer focused business questions, and Power BI to turn the results into a dashboard that is easier to explore. The main takeaway is that looking at revenue alone is not enough. Purchase frequency, discounts, customer groups, product ratings, and subscription behavior provide additional context for better business decisions.